

Article

Analysis and Prediction of Traffic Conditions Using Machine Learning Models on Ikorodu Road in Lagos State, Nigeria

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Abstract: Traffic counts are essential for assessing road capacity to provide efficient, effective, and safe mobility. However, current methods for generating models for traffic count studies are often limited in their accuracy and applicability, which can lead to incorrect or imprecise estimates of traffic volume. This study focused on analyzing and predicting traffic conditions on Ikorodu Road in Lagos State. The analysis involved an examination of historical traffic data, specifically focusing on daily and hourly traffic volumes. The prediction involved the use of machine learning models, including decision trees, gradient boosting, and random forest classifiers. The results of this study revealed significant variations in traffic volume across different days of the week and times of the day, indicating peak and off-peak periods. The study also highlighted the need for a more comprehensive approach that includes additional factors, such as weather conditions, road work, and special events, which could significantly impact traffic volume.

Keywords: road network; traffic flow; vehicle speed; traffic volume; traffic count; machine learning



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1. Introduction

The basis of an economy is the effective movement of people, goods, and services within regions that are covered by transportation networks [1]. However, the interconnectivity of the various roads that make up the network, as well as the efficiency of upstream and downstream vehicular movements, are key components of a successful road network. Consequently, a nation's road system is a crucial part of its infrastructure [2]. A well-designed road network creates effective and sufficient daily traffic flow, which can spur economic expansion. Owing to its impact on inhabitants' daily activities, mobility is crucial to the viability of communities [3]. Transportation systems play an integral role in the movement of people, goods, and services throughout the city, thus supporting socio-economic activities, social interactions, and the overall quality of life [4–10]. The effectiveness of transportation networks depends on several factors, including infrastructure capacity, design, interconnection, and operational efficiency [7,11–17]. These include road networks, which are the main modes of transport in many urban centers worldwide, with road traffic forming the foundation of daily mobility [18–26]. Road infrastructure therefore has a profound impact on urban economic performance, social well-being, and environmental sustainability [26–32]. In many cities worldwide, roads are the main means of transportation, and road traffic is the core of daily mobility. Road networks have a profound impact on urban economic performance, social well-being, and environmental sustainability [33–37]. Nigeria's largest city, Lagos, exemplifies the challenges faced by urban transportation systems.

Over the past few decades, the population has grown exponentially, with estimates of over 20 million people, driven by migration from rural areas. The rapid growth of the city has put immense pressure on infrastructure, especially its transportation system, which has become insufficient to meet growing demands [38,39]. Insufficient road networks, traffic congestion, and inadequate public transportation are some of the most common problems faced by residents daily, highlighting the importance of sustainable urban planning. It is becoming increasingly difficult for Lagos' road networks, which were designed decades ago, to keep up with the growing demand for mobility as the city grows [40–42]. In addition to causing economic and environmental inefficiencies, Lagos' congested roads are a major barrier to efficient transportation. Traffic flow and congestion are critical issues in urban transportation systems, particularly in rapidly growing cities such as Lagos, Nigeria. These challenges significantly impact urban mobility, economic productivity, and residents' quality of life [36,42–44]. Numerous studies [45–47] have identified the causes and consequences of urban congestion, including high traffic volume, insufficient road capacity, and inadequate traffic management. Additional contributing factors include suboptimal public transportation systems, rapid urbanization, and increasing car ownership rates in developing cities [48–53]. Traditional traffic data collection methods, such as manual counting and road sensors, exhibit limitations in capturing the full spectrum of traffic dynamics. These approaches often yield incomplete or inaccurate data, are labor-intensive, and struggle to capture real-time changes in traffic patterns. Moreover, they may not be suitable for large-scale implementation in cities with limited resources or infrastructure [14,54–57].

In recent years, machine learning techniques have emerged as powerful tools for analyzing and predicting traffic conditions. Unlike traditional methods, ML algorithms can process large volumes of data, identify complex patterns, and generate accurate predictions based on historical information. This capability facilitates a more comprehensive understanding of traffic dynamics and enables more effective traffic management strategies [58–64]. Research has demonstrated the efficacy of support vector machines, artificial neural networks, decision trees, and random forests in predicting traffic flow and congestion levels with high accuracy [65–72]. These algorithms can incorporate a wide range of variables, including historical traffic data, weather conditions, time of day, and special events, to generate robust predictions. Some researchers have also explored the potential of deep learning techniques, such as recurrent neural networks and long short-term memory networks, which are particularly well suited for analyzing time-series data like traffic patterns.

The application of machine learning in traffic prediction has significant implications for urban transportation management. Real-time traffic prediction can assist authorities in optimizing signal timing, managing congestion, and improving road safety. By anticipating traffic bottlenecks and congestion hotspots, city planners and traffic managers can implement proactive measures to alleviate congestion before it occurs [60,73–76]. These measures may include dynamic lane management, adaptive traffic signal control, or real-time route guidance for drivers. Predictive models can also support infrastructure planning by forecasting future traffic volumes and identifying potential congestion hotspots. This information is invaluable for urban planners and policymakers in making informed decisions about road network expansions, public transportation investments, and land-use planning [77–80]. By anticipating future traffic patterns, cities can develop more sustainable and efficient transportation systems that can accommodate growing populations and changing mobility needs.

However, challenges persist in the development and deployment of predictive models, including model validation and the need for high-quality, granular data. Ensuring the accuracy and reliability of ML models in diverse urban contexts is crucial for their effective

implementation. Additionally, data privacy and security concerns must be addressed when collecting and utilizing large-scale traffic data. There is a notable lack of research on the application of machine learning for traffic prediction in rapidly urbanizing African cities like Lagos, where data infrastructure is limited and traffic conditions are highly variable. These cities often face unique challenges, including informal transportation systems, rapidly changing urban landscapes, and limited technological resources. Developing effective ML-based traffic prediction models in these contexts requires innovative approaches to data collection and model development.

This study aims to address this gap by applying supervised machine learning techniques to predict traffic conditions on Ikorodu Road in Lagos. By leveraging video-based data collection methods and ML algorithms, the research seeks to provide accurate and real-time traffic predictions in a challenging urban environment. This approach has the potential to overcome some of the limitations of traditional data collection methods and provide a more comprehensive understanding of traffic dynamics in Lagos. The research contributes to the broader field of intelligent transportation systems (ITSs) and smart city initiatives. By demonstrating the feasibility and effectiveness of ML-based traffic prediction in a complex urban environment like Lagos, this study could pave the way for similar applications in other rapidly growing cities in Africa and beyond. The insights gained from this research could inform the development of more effective traffic management strategies, improve urban mobility, and ultimately enhance the quality of life for urban residents.

2. Literature Review

2.1. Traffic Flow and Congestion

Traffic flow refers to the movement of vehicles within a transportation network and is influenced by a variety of factors including traffic volume, road conditions, vehicle types, driver behavior, and environmental conditions. It is essential to understand traffic flow dynamics to develop strategies to improve the efficiency of road networks [53,57,81–83]. According to [84], traffic flow can be categorized into different regimes such as free flow, synchronized flow, and jam flow. In free-flow conditions, vehicles move smoothly with minimal interaction between them, whereas in synchronized flow, vehicles move at slower speeds but maintain a coordinated pattern. Traffic congestion occurs when road capacity is exceeded, leading to slower speeds, increased delays, and stop-and-go conditions [85].

Several studies have highlighted the causes and consequences of urban congestion. For instance, ref. [86] demonstrated that the primary causes of congestion include high traffic volume, insufficient road capacity, and poor traffic management. Similarly, ref. [87] suggested that traffic accidents, a lack of driver awareness, and a poor understanding of traffic signs contribute to accidents and congestion in Nigerian cities. A study focused on urban traffic congestion [88] emphasized the importance of adopting comprehensive traffic management strategies that consider the unique characteristics of local traffic patterns.

2.2. Traditional Traffic Data Collection Methods

Traditionally, traffic data have been collected through manual counting, road sensors, and cameras. These methods have been widely used for decades to monitor vehicle counts and determine traffic volumes at different times of day [89]. Manual counting involves human observers recording the number of vehicles passing a particular point on the road, whereas automated methods use sensors, such as inductive loops or infrared sensors, to detect vehicles and gather data on traffic conditions [90].

However, these methods also have several limitations. For example, manual counting is labor-intensive, prone to human error, and can only be performed at specific locations and times. Automated sensors, on the other hand, are often costly to install and

maintain, and their data collection capabilities are limited to specific points on the road network [82,84,89,91]. As such, these methods do not capture the full range of traffic dynamics such as vehicle interactions, congestion patterns, or the impact of external factors such as weather or accidents.

2.3. Machine Learning in Traffic Prediction

In recent years, machine learning (ML) techniques have emerged as powerful tools for analyzing and predicting traffic conditions. Unlike traditional methods, ML algorithms can process large volumes of traffic data, identify complex patterns, and provide accurate predictions based on historical data. The application of ML in transportation research has grown significantly, as evidenced by several studies exploring its use in traffic flow prediction, congestion forecasting, and optimization of traffic management systems [53,92,93].

Machine learning models have the potential to revolutionize traffic prediction by learning from historical traffic data and adapting to changing traffic conditions [94]. In [95], the authors demonstrated the use of support vector machines (SVMs) to accurately predict traffic flow on urban roads, traffic volumes, and congestion levels, outperforming traditional models based on linear regression. Other ML techniques, such as artificial neural networks (ANNs), decision trees, and random forests, have also been widely applied for traffic prediction [90,92,94–96]. Ref. [97] employed artificial neural networks to model traffic flows and predict highway congestion. Their model could forecast traffic volumes and congestion patterns with a high degree of accuracy, even during periods of heavy traffic. Similarly, ref. [98–101] used random forests to predict traffic conditions at various intersections in a busy urban area, achieving notable improvements in prediction accuracy compared to conventional methods.

Several studies have explored the integration of multiple data sources for more comprehensive traffic prediction models [70,95,102–104]. For instance, ref. [97] combined real-time traffic data, weather information, and historical traffic patterns to develop a machine learning model that predicts traffic conditions across a city. The integration of diverse data sources has proven to enhance the accuracy of traffic predictions, particularly in complex urban environments such as Lagos [105,106].

2.4. Applications on Machine Learning in Urban Transportation Systems

The use of machine learning in traffic prediction has significant implications for urban transportation management. Real-time traffic prediction can assist traffic authorities in optimizing traffic signal timing, managing congestion, and improving overall road safety [64,90,93,94,104,106]. Additionally, predictive models can support infrastructure planning by forecasting future traffic volumes and identifying potential congestion hotspots that require road expansions or upgrades [107]. In the context of Lagos, where traffic congestion is a major concern, machine learning models can help policymakers allocate resources more effectively and prioritize investments in transportation infrastructure. For example, by predicting peak traffic periods, authorities can adjust traffic signal timings, implement variable-toll pricing, or deploy traffic officers to manage congestion at critical intersections [72,108,109]. Predictive traffic models can inform public transportation planning by identifying areas with a high demand for buses or other modes of transport. [101,110–112].

2.5. Challenges in Traffic Prediction and Model Validation

Despite promising applications of machine learning in traffic prediction, several challenges remain in the development and deployment of predictive models. One of the primary challenges is model validation [104,109,113]. According to [114], the calibration and validation of traffic prediction models can be complex and time-consuming. This is

particularly true when using data from different sources because discrepancies in data quality or coverage can lead to inaccurate predictions. In addition, traffic conditions can be highly variable and influenced by numerous external factors, such as accidents, road-work, and weather conditions, making it difficult to develop models that are universally applicable across different scenarios [115,116].

Another challenge is the requirement for high-quality granular data. For machine-learning models to produce accurate predictions, they must be trained on a large dataset that captures the full range of traffic conditions. However, in many cities, there is a lack of sufficient traffic data, especially in developing countries such as Nigeria, where traffic counting and monitoring infrastructure are often inadequate [117–120]. This limitation can hinder the development of effective machine learning models and limit their potential for real-world applications [121–123].

Although significant progress has been made in the use of machine learning for traffic prediction, there remains a need for more research on its application in rapidly urbanizing cities, particularly in African countries, such as Nigeria. Existing studies have tended to focus on developed regions, where the data infrastructure is more robust and traffic conditions are often more predictable. In contrast, cities such as Lagos present unique challenges, including highly variable traffic conditions, limited data availability, and rapid urbanization [124–127]. There is also a lack of studies that specifically focus on the application of machine learning to traffic prediction on major urban roads in Lagos, such as Ikorodu Road. This gap in the literature presents an opportunity for further research, as understanding the traffic dynamics on such roads is critical for improving the overall efficiency and safety of a city's transportation system [127,128].

3. Materials and Methods

In this study, we aimed to analyze and predict traffic conditions on Ikorodu Road, one of Lagos' busiest highways, using machine learning models. The methodology was designed to provide insights into how traffic flows and congestion in this highly urbanized environment can be better managed. Below, we describe the steps taken, including data collection, preprocessing, model development, and the evaluation of model performance.

3.1. Study Area

Ikorodu Road is a critical transportation route in Lagos that connects the northern parts of the city to the central business districts, specifically between the Jibowu and Fadeyi bus stops, located at 6.5178° N, 3.3679° E. This area is situated within a bustling urban environment and is characterized by high volumes of traffic. A variety of data collection techniques and tools, including traffic sensors and cameras, will be employed to gather information on the traffic volume and flow patterns in the study area. The collected data will serve as the basis for developing precise and comprehensive geometric models of highways and their adjacent infrastructures.

3.2. Methods

To obtain a comprehensive understanding of traffic patterns on Ikorodu Road in Lagos, Nigeria, data were collected on all days of the week, from Monday to Sunday. The methodology involved traffic volume data acquisition and preprocessing, followed by machine learning (ML) model development and the evaluation for traffic classification. A summary of the research process for the analysis and prediction of traffic is displayed in Figure 1, with the dataset preprocessing and model architecture illustrated in Figure 2.

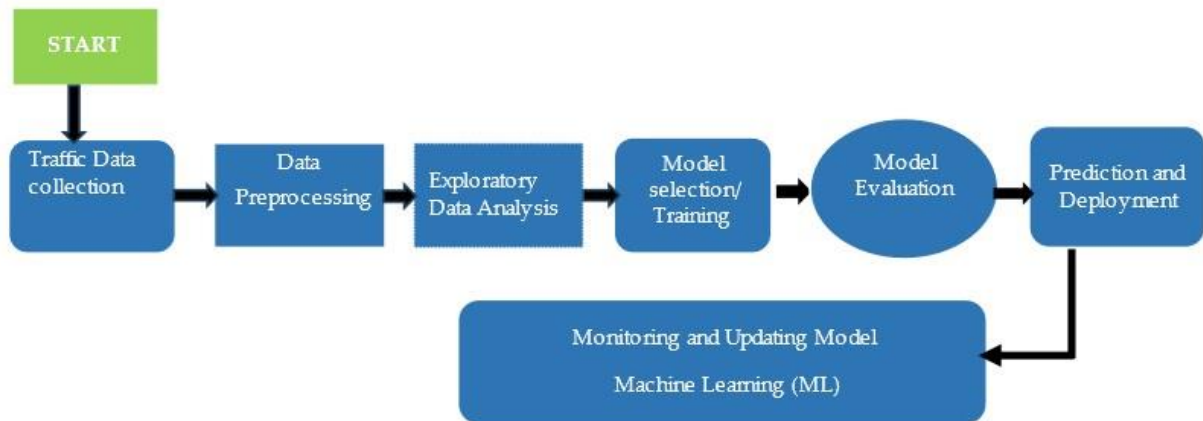


Figure 1. Flow chart on the analysis and prediction of traffic.

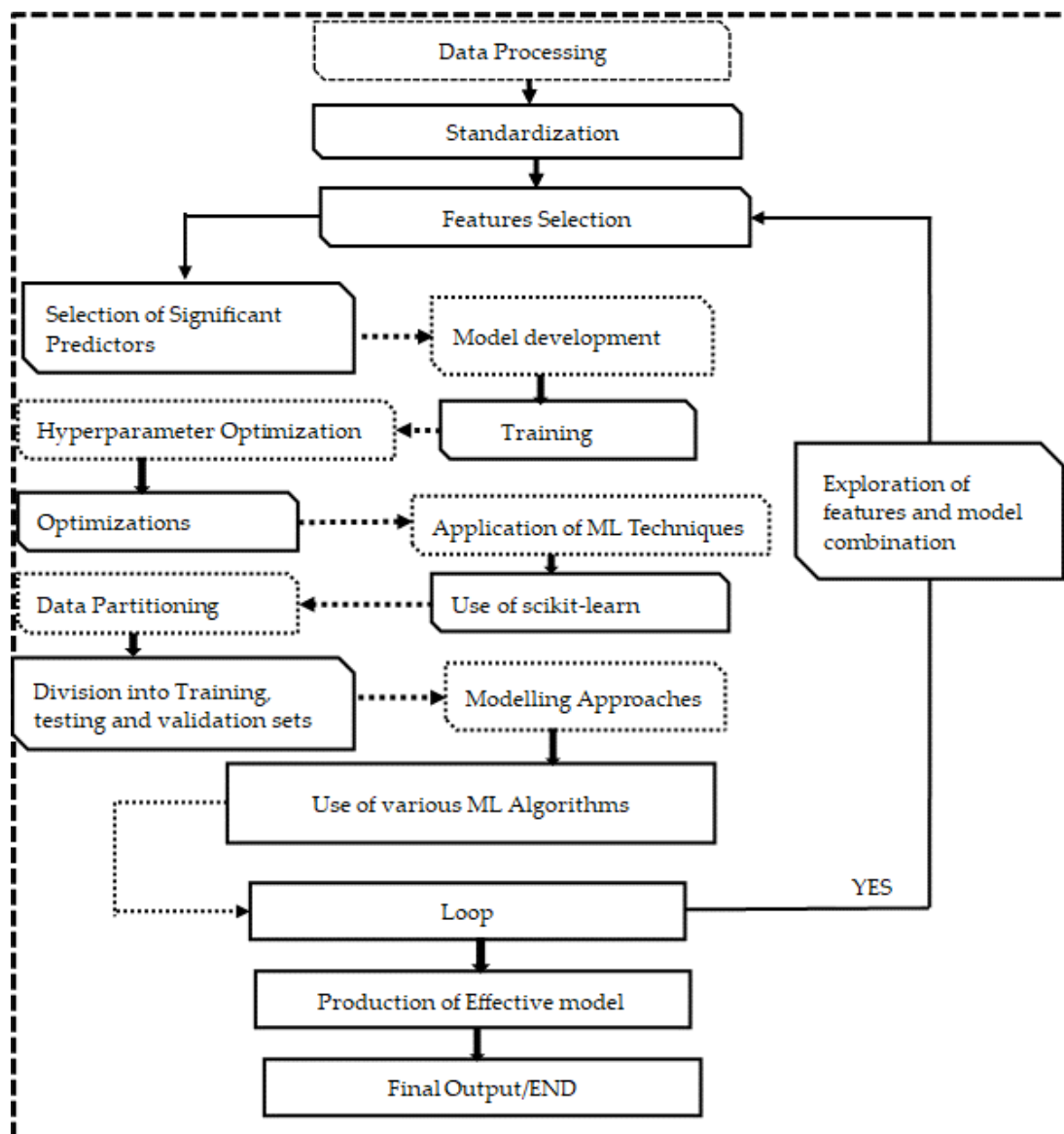


Figure 2. Flow diagram illustrating dataset preprocessing and machine learning model architecture.

3.3. Traffic Count Survey

The traffic volume data were collected using a combination of manual and automated methods. Video cameras were strategically installed along Ikorodu Road between Jibowu and Fadeyi bus stops to capture continuous traffic flow over a full week. The use of pre-recorded video footage allowed for flexibility in data analysis as well as improved accuracy compared to on-site manual counts, which are often limited by weather conditions, human error, and resource intensity. Vehicles were counted at 1 h intervals for each 24 h cycle, and traffic was classified into five categories: cars, jeeps, buses, trucks, and tricycles. The resulting dataset was organized in a structured format (CSV or spreadsheet) with variables representing the day of the week, time interval, vehicle type, and hourly vehicle count. To ensure reliability, the dataset was subjected to quality control to detect and correct anomalies or missing values. The Average Daily Traffic (ADT) was calculated for each day and the Annual Average Daily Traffic (AADT) was derived by dividing the total yearly traffic counts by 365. These metrics were used to assess the current traffic demand and identify infrastructure needs.

Manual video-based counting, although accurate, is time-consuming and can take up to three hours to analyze one hour of footage, depending on factors such as traffic density and video quality. However, it remains the preferred method when automated systems are not available or are not practical. The accuracy of such counts is vital for planning studies including delay analysis, capacity estimation, and traffic impact assessments. Manual traffic counts were conducted using pre-recorded video footage captured using strategically positioned cameras. This method was selected over on-site manual counting because of its greater reliability in adverse weather, reduced labor intensity, and increased accuracy. As noted in previous studies [129,130], video-based traffic counts offer a cost-effective and scalable solution for urban traffic analysis, although challenges such as human errors, video quality, and survey design inefficiencies may still arise [97].

Traffic was recorded at hourly intervals for 24 h each day. Vehicles were classified into five main categories: cars, jeeps, buses, trucks, and tricycles. The data were structured in tabular format (CSV), with fields for the day of the week, hour, vehicle type, and vehicle count. Data quality checks were performed to identify and rectify the anomalies, missing values, and inconsistencies. The Average Daily Traffic (ADT) was computed from the recorded volumes. The Annual Average Daily Traffic (AADT) was derived by aggregating the daily traffic counts over the year and dividing it by 365. This metric serves as a basis for estimating the current service demand and identifying areas requiring infrastructural improvements.

3.4. Machine Learning Models (ML)

ML is an artificial intelligence (AI) branch that aims to develop statistical models to improve computer system performance by learning data [131,132] including supervision, unsupervised learning, and reinforcement. Although supervision and reinforcement learning algorithms can involve human supervision, non-supervised learning algebra does not depend on label data or human guidance. Our study used supervised ML algorithms to classify traffic numbers and volumes using various vehicle datasets. We used vehicle training datasets and trained three different ML algorithms: decision trees, gradient increase, and random forest classification systems, and tested their performance using R programming languages [131].

After feature selection, classification models were developed to predict traffic conditions. These models classify traffic into three categories: 'Low', 'Medium', and 'High'. These categories represent different levels of traffic volume. Three different classification models were used: the decision tree classifier (DT), gradient boosting classifier (GBC),

and random forest classifier (RFC) [133–135]. These models were chosen because of their robustness and performance with classification tasks, especially with multi-class problems, such as ours. The models were trained using the training dataset and selected features. The performances of the models were evaluated using cross-validation scores, and the best-performing model was selected based on the average score [136].

3.4.1. Random Forest Model (RFC)

Random forests are widely used as comprehensive learning algorithms for classification and regression. This algorithm uses multiple decision trees to improve the accuracy and robustness of models. Unlike individual decision trees, random forests are less likely to be overfitted because they bind multiple trees with different biases and differences. Additionally, they can efficiently handle high-dimensional data with many functions by randomly selecting a set of functions for each tree. As a result, algorithms can handle large and complex datasets [85]. We used random forest algorithms to train models using random forest packages [137] in R programming languages. We refined the model's hyperparameters using Bayesian optimization techniques based on models, including the random sampling of variables at each step of a decision tree, the minimum size of an internal node, and the number of decision trees. The performance of the model was evaluated using cross-validation, and the optimal values of the hyperparameters were selected based on their best performance.

3.4.2. Decision Tree Model (DT)

Decision trees (DTs) are widely used machine learning algorithms that can be applied to classification and regression problems. This non-parametric algorithm can handle large-scale complex datasets without forcing rigid parametric structures and is a versatile tool for different applications [138]. The DT algorithm creates a tree model, with the internal node of the tree representing an input feature decision and each leaf node representing a class label or target value. DT models are suitable for multilevel classification problems because they capture the nonlinear relationship between input characteristics and target variables [139,140].

3.4.3. Gradient Boosting Classifier (GBC)

Gradient boosting classifiers (GBC) are highly accurate, efficient, and popular. This is an ensemble method that combines several weak learners, such as decimal trees, into a single strong learner [140]. The gradient boost package is typically used to implement regression models in R. The package also provides support for the modification of high-performance parameters, which can significantly improve the performance of the model. The hyperparameters provide several options for adjustment, including learning rates, the maximum depth of each tree, the number of trees, and regularization parameters (alpha and gamma). Bayesian optimization can be used to increase the hyperparameters in gradient boosting.

3.5. Reliability Analysis

Before developing the prediction models, a comprehensive statistical analysis of the collected traffic data was conducted. The main objectives were to understand the underlying structure of the data, identify patterns, and extract meaningful insights. The following statistical methods were used.

3.5.1. Descriptive Statistics

Descriptive statistics provided an overview of the data, including central tendencies (mean) and measures of dispersion (variance and standard deviation). This helped us understand the general trends and variability in traffic volume.

Mean: The mean of a dataset X with n elements is given by the following equation:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Variance: The variance of dataset X with n elements is given by the following equation:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \quad (2)$$

Standard Deviation: The standard deviation is the square root of the variance.

3.5.2. Correlation Analysis

Correlation analysis was conducted to discover potential relationships between different variables in the dataset. This helped our understanding of how variables such as the time of day and the type of vehicle influenced traffic volume.

Pearson's correlation coefficient between two variables X and Y is given as follows:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)}{\sqrt{\sum_{i=1}^n (x_i - \mu_x)^2 \sum_{i=1}^n (y_i - \mu_y)^2}} \quad (3)$$

3.5.3. Model Development

During preprocessing, the standardization process was divided into three sets: training, testing, and validation. A feature selection model was used to pinpoint the most significant predictors of traffic volume from the training and validation datasets, which [113,116,141] is characterized by a loop connecting the feature selection, modeling, and hyperparameter optimization modules. This loop automatically explores different combinations of features and modeling methods, ultimately producing the most effective model with the optimal subset of features based on the input data [142].

3.5.4. Data Preprocessing and Partitioning

In the initial phase of data processing, numerical values were assigned to each independent variable. These values were then standardized to follow a normal distribution (0, 1) [93,141,143].

3.5.5. Feature Selection

Feature selection was performed using standardized and partitioned data. Function S was used to determine whether a feature column should be selected in a binary manner. The selected data were then modeled using both linear and nonlinear methods. During inference, the result for a given data point is computed as follows, where F represents the trained model, and X_i represents the features [41,48,50,141].

$$Y_i = F(X_i S(X_i)) \quad (4)$$

3.5.6. Model Evaluation and Validation

The performance of the models was evaluated using a range of techniques. Cross-validation was used during the model selection phase to ensure that the chosen model did not overfit the training data and could generalize well to the unseen data. The

cross-validation process involved splitting the training data into five subsets or ‘folds’. Four of these folds were used for training, and the remaining fold was used for testing. This process was repeated five times, and each fold was used once for testing [64,93]. The results of each run were averaged to obtain the final cross-validation scores for the model.

The ML models are presented and discussed using a confusion matrix and various performance metrics, such as overall accuracy, sensitivity (ability to detect positive instances), specificity (ability to detect negative instances), negative predicted value (NPV), positive predicted value (PPV), and balanced accuracy (the average of sensitivity and specificity) [101,109,113]. Owing to the imbalanced dataset used for training and testing purposes, additional informative performance metrics, such as precision, recall, and F1_score, were utilized to assess the efficacy of the ML models.

$$\text{Overall Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (5)$$

where TP = true positive (correctly predicted as positive), TN = true negative (correctly predicted as negative), FP = false positive (incorrectly predicted as positive), and FN = false negative (incorrectly predicted as negative).

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (6)$$

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) \quad (7)$$

$$\text{F1_score} = \frac{2 \times \text{precision} \times \text{Sensitivity}}{\text{precision} + \text{Sensitivity}} \quad (8)$$

In addition, a classification report was generated for each model. This report included metrics such as the precision, recall, and F1 score for each traffic condition (‘High’, ‘Medium’, and ‘Low’). These metrics provided detailed insights into the performance of the models. The model was validated by using a validation dataset. The results of this validation process provided the final assurance of the model’s predictive power and ability to generalize new data.

3.5.7. Predicting Traffic Conditions

The chosen model can be used to predict traffic conditions based on the features selected during the feature selection process. To make a prediction, the user must input the feature values into the model. The model then outputs a prediction of the traffic condition (‘High’, ‘Medium’, or ‘Low’). These predictive models are powerful tools for traffic management. They can be used to predict future traffic conditions based on current and historical data, allowing proactive measures to be taken to manage and control traffic flows [109,124].

4. Results and Discussion

The results of the traffic data analysis were conducted on Ikorodu (Jibowu Section) Road. The analysis was performed using the outlined methods, including traffic count analysis, time-based analysis, statistical analysis, and the development of predictive models. The primary objective of these methods is to gain an in-depth understanding of the traffic patterns along the studied segment of the road and to predict future traffic conditions. Such insights can aid effective traffic management and planning.

The Results Section provides insights into the traffic volume observed daily and at different times of the day. Subsequently, the outcomes of the statistical analyses are presented, which include the mean, variance, and standard deviation of the traffic volume

for each day. Furthermore, a correlation analysis was conducted to examine the relationship between different variables, such as the time of day, day of the week, and traffic volume.

The performance of the predictive models developed in this study was then examined, highlighting their accuracy in classifying traffic into various categories: ‘Low’, ‘Medium’, and ‘High’. The implications of these models for predicting future traffic conditions were also explored. The analysis and interpretation of these results are integral to achieving the research objectives. They provide valuable insights into the traffic patterns on Ikorodu Road and their ability to forecast future traffic conditions. These findings are critical for the development of effective traffic management strategies and solutions.

4.1. Daily Traffic Volume

An examination of the traffic volume for each day of the week provided crucial insights into the traffic patterns on Ikorodu Road. As illustrated in Table 1, the total flow varied significantly across days, indicating different traffic dynamics at play. The highest total flow was observed on Monday, with a traffic volume of approximately 5582 vehicles per day. This was closely followed by Tuesday, with an estimated 5372 vehicles per day. Conversely, Saturday experienced the lowest total flow, with approximately 3083 vehicles per day. The analysis of the average flow per hour revealed similar patterns. The average flow was the highest on Monday (465.17 vehicles per day) and the lowest on Saturday (256.92 vehicles per day).

Table 1. Traffic volume for each day of the week.

Day	Total Flow (Veh/day)	Average Flow (Veh/day)
Monday	5582	465.17
Tuesday	5372	447.67
Wednesday	4739	394.92
Thursday	4996	416.33
Friday	5239	436.58
Saturday	3083	256.92
Sunday	3832	319.33

The traffic volume on Ikorodu Road fluctuates throughout the week, reflecting varying patterns of commuter activity. Figure 3 illustrates these patterns using confidence and prediction bands, emphasizing the variability and predictability in traffic flow. Monday is the busiest day with 5582 vehicles, as people return to their workweek routines, followed by Tuesday with 5372 vehicles. Wednesday saw a slight drop to 4739 vehicles, reflecting a mid-week lull, while Thursday and Friday saw increased traffic again, reaching 4996 and 5239 vehicles, respectively, as people wrap up their week and prepare for the weekend. The weekend shows a stark contrast, with Saturday having the lowest traffic of 3083 vehicles, as many people stay home or engage in non-work activities. Sunday saw a moderate increase to 3832 vehicles, which is still significantly lower than on weekdays. In summary, weekdays, particularly Mondays and Fridays, experience the heaviest traffic due to commuting and work-related activities, while Saturdays and Sundays see reduced traffic, reflecting the shift to leisure and rest. Understanding these patterns is crucial for improving traffic management and planning.

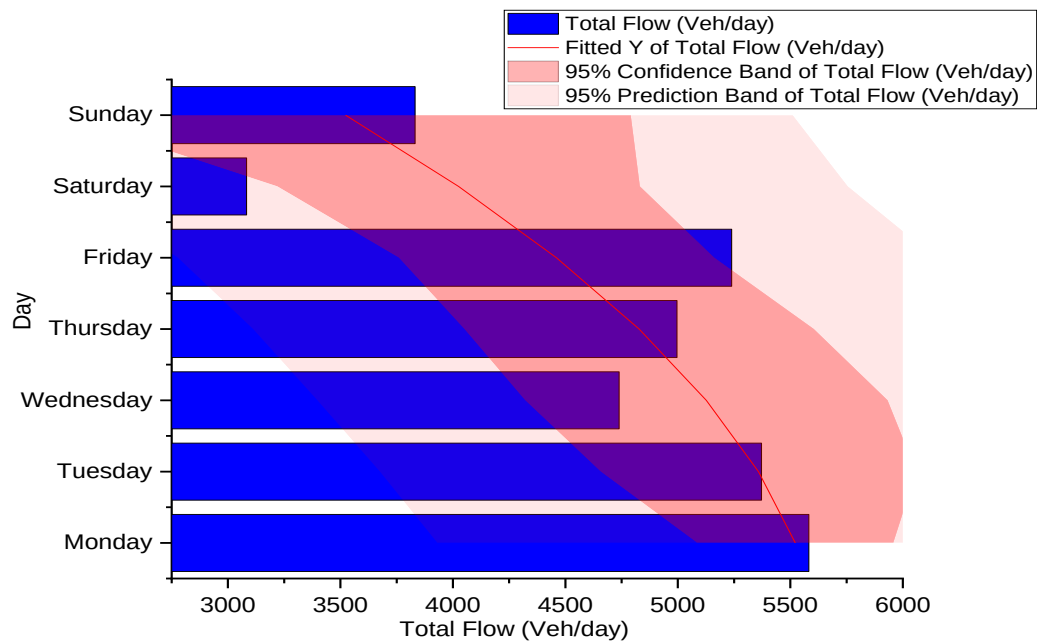


Figure 3. Confidence and prediction band in relationship with the traffic volume of vehicle.

4.2. Time-Based Traffic Volume

Analyzing traffic volume in relation to time is essential, as it helps determine peak and off-peak hours, which could be crucial in planning and managing traffic flow.

Table 2 and Figure 4 below is a summary of the total traffic flow for each type of vehicle per hour.

Table 2. Total traffic flow per hour.

Time	Car Total	Jeeps Total	Trucks Total	Buses Total	Tricycles Total	Bicycles Total
6:00	2258	1141	34	270	211	12
7:00	2271	590	23	455	174	7
8:00	1145	268	17	266	187	12
9:00	1173	276	15	313	269	6
10:00	733	154	15	186	294	10
11:00	438	175	12	309	187	15
12:00	471	234	24	262	195	9
13:00	459	241	9	160	168	6
14:00	1245	295	10	197	166	13
15:00	1863	450	23	232	170	5
16:00	2730	721	19	431	195	9
17:00	3445	555	14	321	167	10
18:00	2258	1141	34	270	211	12
19:00	2271	590	23	455	174	7
20:00	1145	268	17	266	187	12
21:00	1173	276	15	313	269	6
22:00	733	154	15	186	294	10
23:00	438	175	12	309	187	15
0:00	471	234	24	262	195	9
1:00	459	241	9	160	168	6
2:00	1245	295	10	197	166	13
3:00	1863	450	23	232	170	5
4:00	2730	721	19	431	195	9
5:00	3445	555	14	321	167	10

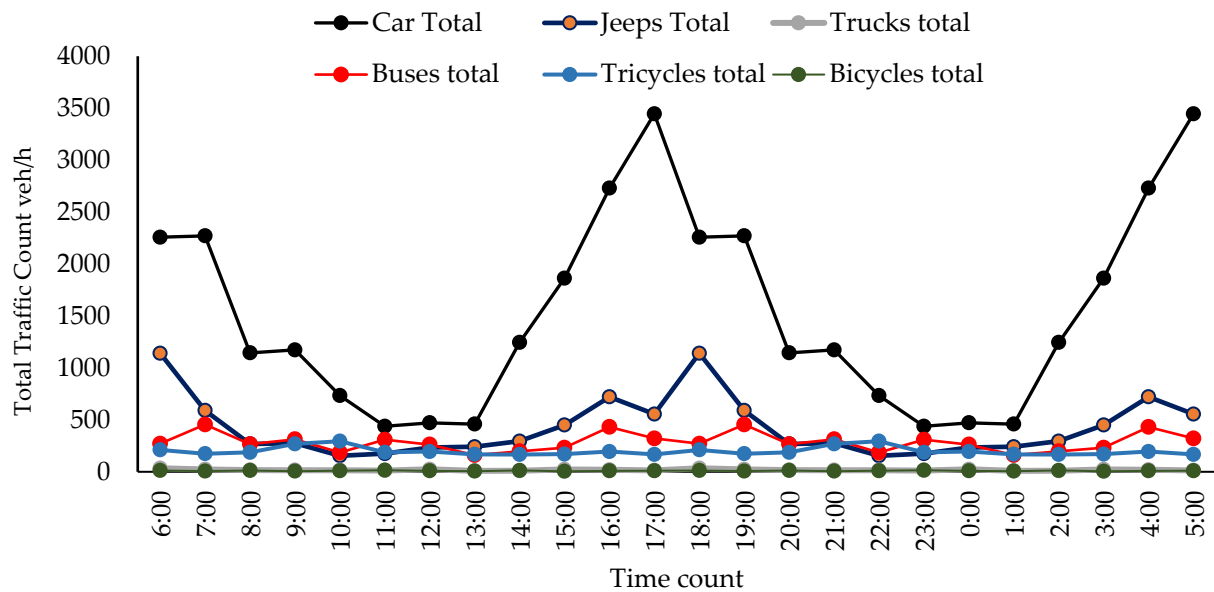


Figure 4. Total traffic count by day and vehicle.

The traffic data for Ikorodu Road revealed significant variations in vehicle flow during different hours of the day. The highest traffic volume was observed between 16:00 (4:00 p.m.) and 17:00 (5:00 p.m.), with a peak of 3445 vehicles, predominantly composed of cars (555), jeeps (721), and buses (431). These hours align with the evening rush period, indicating a surge in commuter and transportation activity. Conversely, the lowest traffic volumes were recorded at 12:00 p.m. and 1:00 a.m., when only 438 vehicles were observed, reflecting a lull in road usage during midday and late-night hours. Throughout the day, cars consistently represented the largest proportion of traffic on Table 3 and Figure 5, with their numbers peaking in the afternoon. Jeeps also maintained a substantial presence, particularly during late morning and afternoon periods. Buses showed notable spikes at 6:00 a.m. and 17:00 p.m. in Figure 6, reflecting the role of public transportation during the morning and evening rush hours. Trucks, although less frequent, exhibited higher counts in the early morning and lower numbers overnight. Tricycles and bicycles accounted for a smaller fraction of the total traffic, but remained relatively stable in volume, with tricycles peaking at 6:00 a.m. and bicycles showing consistent but low numbers throughout the day. The traffic patterns on Ikorodu Road exhibited peak congestion during the late afternoon and early evening, predominantly driven by cars, jeeps, and buses. In contrast, the traffic volume was significantly lower during the late night and midday periods. These insights are essential for traffic management, urban planning, and optimization of road infrastructure to accommodate peak and off-peak traffic conditions effectively.

Understanding these traffic volume trends across different times of the day is crucial for making informed decisions regarding traffic planning and management. For instance, during peak traffic hours, traffic control measures may need to be intensified, and roads may be designed with a higher capacity. Conversely, resources can be allocated to others during off-peak hours. Infrastructure planning should also consider these patterns to address traffic congestion efficiently.

Table 3. Total traffic flow and average traffic flow per hour.

Time	Car Total	Cars Average	Jeeps Total	Jeeps Average	Trucks Total	Trucks Average	Buses Total	Buses Average	Tricycles Total	Tricycles Average	Bicycles Total	Bicycle Average
6:00	2258	322.57	1141	163	34	4.86	270	38.57	211	30.14	12	1.71
7:00	2271	324.43	590	84.29	23	3.29	455	65	174	24.86	7	1
8:00	1145	163.57	268	38.29	17	2.43	266	38	187	26.71	12	1.71
9:00	1173	167.57	276	39.43	15	2.14	313	44.71	269	38.43	6	0.86
10:00	733	104.71	154	22	15	2.14	186	26.57	294	42	10	1.43
11:00	438	62.57	175	25	12	1.71	309	44.14	187	26.71	15	2.14
12:00	471	67.29	234	33.43	24	3.43	262	37.43	195	27.86	9	1.29
13:00	459	65.57	241	34.43	9	1.29	160	22.86	168	24	6	0.86
14:00	1245	177.86	295	42.14	10	1.43	197	28.14	166	23.71	13	1.86
15:00	1863	266.14	450	64.29	23	3.29	232	33.14	170	24.29	5	0.71
16:00	2730	390	721	103	19	2.71	431	61.57	195	27.86	9	1.29
17:00	3445	492.14	555	79.29	14	2	321	45.86	167	23.86	10	1.43
18:00	2258	322.57	1141	163	34	4.86	270	38.57	211	30.14	12	1.71
19:00	2271	324.43	590	84.29	23	3.29	455	65	174	24.86	7	1
20:00	1145	163.57	268	38.29	17	2.43	266	38	187	26.71	12	1.71
21:00	1173	167.57	276	39.43	15	2.14	313	44.71	269	38.43	6	0.86
22:00	733	104.71	154	22	15	2.14	186	26.57	294	42	10	1.43
23:00	438	62.57	175	25	12	1.71	309	44.14	187	26.71	15	2.14
0:00	471	67.29	234	33.43	24	3.43	262	37.43	195	27.86	9	1.29
1:00	459	65.57	241	34.43	9	1.29	160	22.86	168	24	6	0.86
2:00	1245	177.86	295	42.14	10	1.43	197	28.14	166	23.71	13	1.86
3:00	1863	266.14	450	64.29	23	3.29	232	33.14	170	24.29	5	0.71
4:00	2730	390	721	103	19	2.71	431	61.57	195	27.86	9	1.29
5:00	3445	492.14	555	79.29	14	2	321	45.86	167	23.86	10	1.43

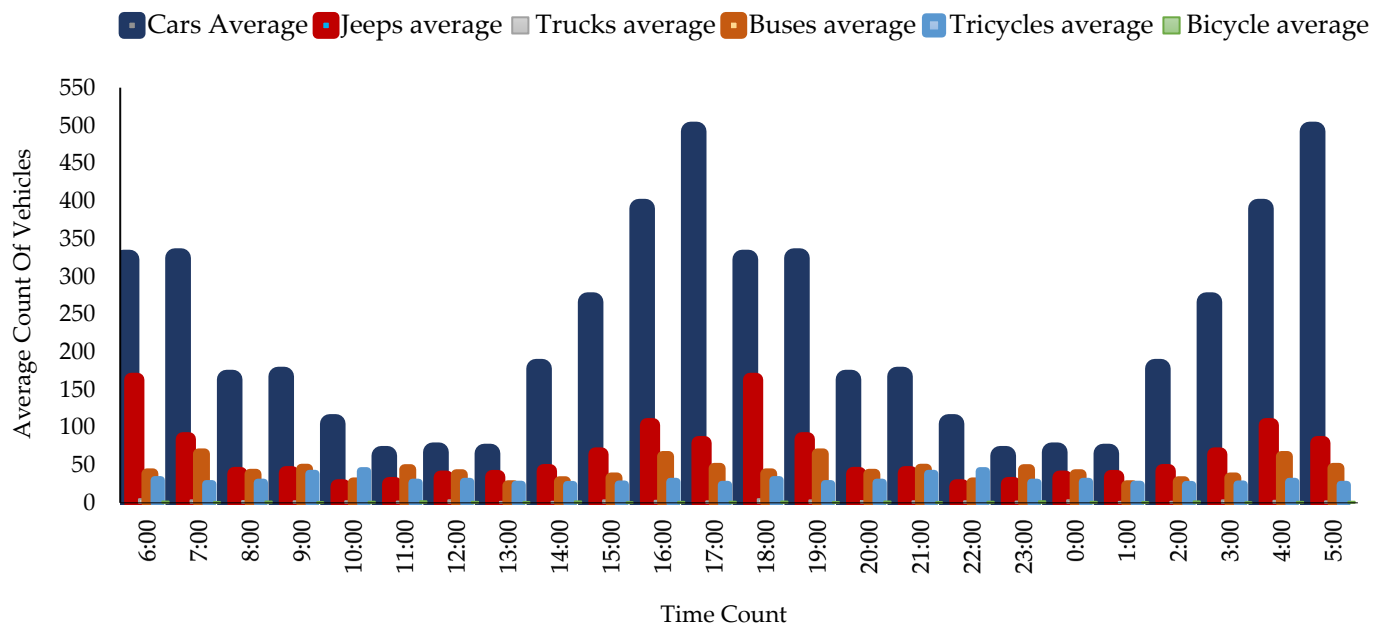


Figure 5. Average traffic count by day and vehicle.

4.3. Descriptive Statistics Results

It was observed that the mean traffic volume tended to decrease from Monday to Wednesday, increased slightly on Thursday and Friday, and then significantly decreased over the weekend on Saturday and Sunday. This may suggest that the road experiences heavier traffic during weekdays than weekends, possibly due to work-related commuters.

The variance and standard deviation, which measure the spread of the traffic volume, are higher during weekdays than during weekends. This might indicate greater variability in traffic volume during weekdays. This variability could be due to various factors, such as differences in work schedules, school runs, and other weekday-specific activities.

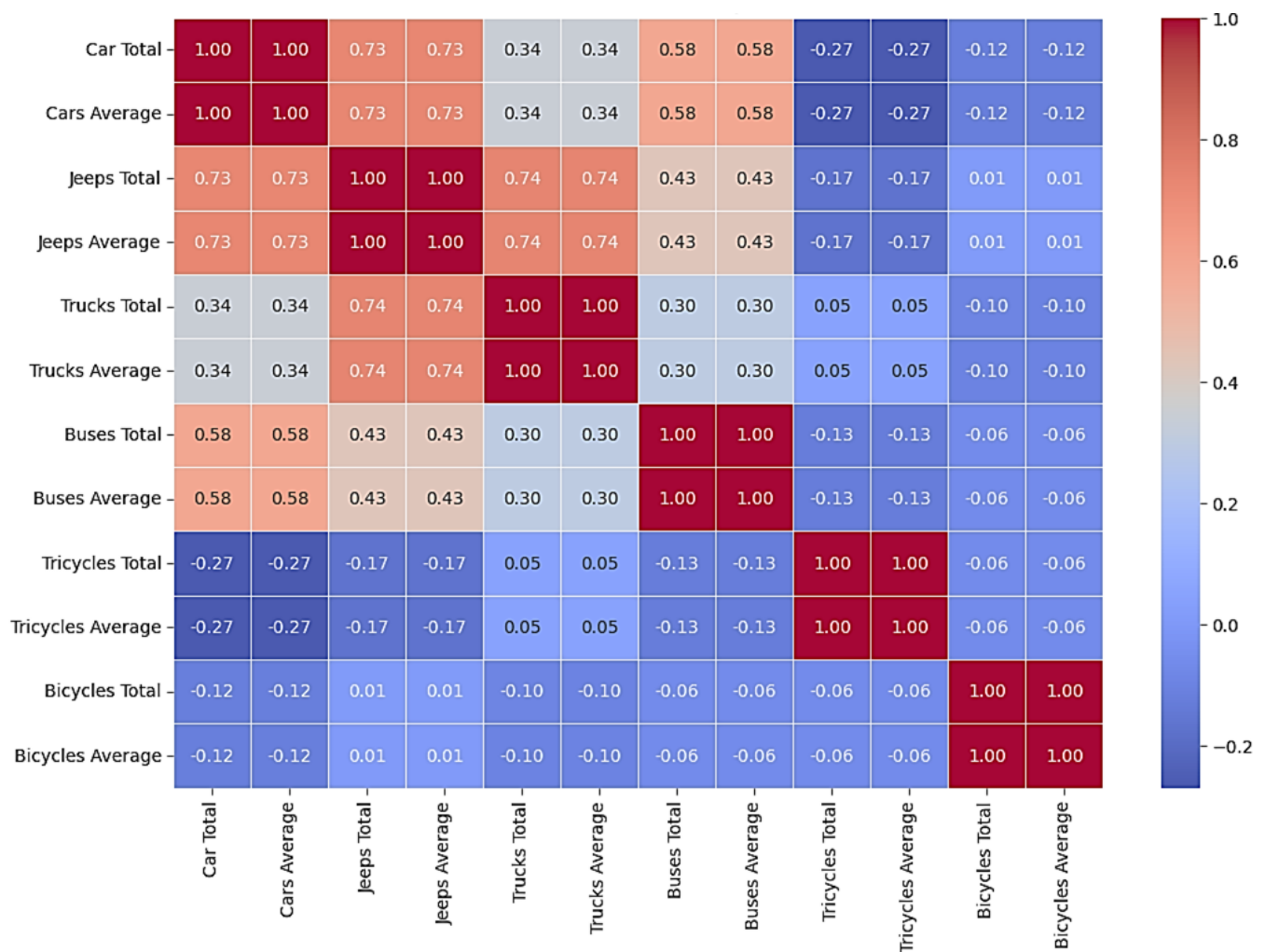


Figure 6. Total and average matrix correlation for traffic flow.

4.4. Correlation Analysis Results

Regarding the correlation result, a correlation value of 0.1048 between the time of day and traffic volume indicated a weak positive correlation as shown in Table 4. In other words, as the day progressed, there was a slightly higher likelihood of increased traffic volume, but this relationship was not very strong.

Table 4. Correlation analysis results.

Variables	Correlation
Time of day and traffic volume	0.1048

4.5. Model Performance Cross-Validation Results

The study incorporated three predictive models: the decision tree classifier, gradient boosting classifier, and random forest classifier. The performance of each model was evaluated using cross-validation and classification. The decision tree classifier achieved an average cross-validation score of approximately 0.881. The gradient boosting classifier slightly outperformed the decision tree classifier, with an average cross-validation score of approximately 0.925. The random forest classifier exhibited an average cross-validation score of approximately 0.894. Although the cross-validation scores were relatively close, the gradient boosting classifier scored the highest, indicating that it had the best general performance in predicting test data during training. The classification reports provided

more detailed information on the performance of each model. Both the decision tree classifier and gradient boosting classifier achieved an accuracy of 0.88 on the test set, while the random forest classifier achieved a perfect accuracy of 1.00. This demonstrates that the random forest classifier had the highest precision, recall, and F1-score for all classes (High, medium, and Low), making it the best-performing model out of the three when tested on unseen data.

The traffic patterns observed on Ikorodu Road throughout the day revealed fluctuations in vehicle flow as illustrated in Table 5 and Figure 7 above, which can be compared to the performance of different machine learning models in a predictive context. Just as traffic volumes rise and fall during the day based on various factors, such as work hours, leisure activities, and commuting patterns, machine learning models decision tree classifiers, gradient boosting classifiers, and random forest classifiers demonstrate different levels of accuracy and efficiency in handling data.

Table 5. Model performance cross-validation results.

Model	Cross-Validation Score	Test Accuracy
Decision Tree Classifier	0.881	0.88
Gradient Boosting Classifier	0.925	0.88
Random Forest Classifier	0.894	1.00

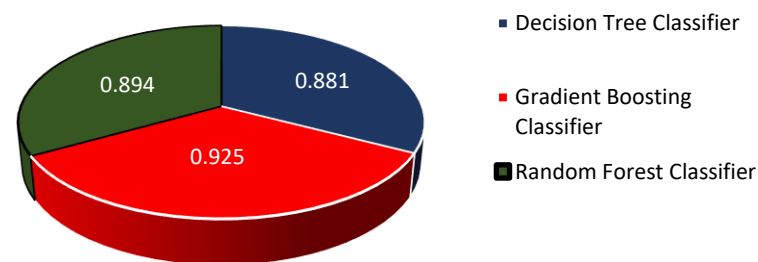


Figure 7. Model performance cross-validation.

The decision tree classifier mirrors predictable traffic patterns. It performs consistently with a cross-validation score of 0.881 and a test accuracy of 0.88 as shown in Table 6 below, similar to how traffic on Ikorodu Road is relatively stable during rush hours, such as the 6:00 a.m. and 17:00 p.m. periods, when car and jeep volume peaks. These periods follow predictable routines in which commuters' behaviors are well understood, much like how decision trees split data into clear categories. However, while the model performs well, it does not fully capture all possible fluctuations, akin to how traffic patterns might remain consistent but occasionally miss variations in flow.

Table 6. Decision tree classifier.

Class	Precision	Recall	F1-Score
High	1.00	1.00	1.00
Low	0.80	0.80	0.80
Medium	0.67	0.67	0.67
Average/Total	0.88	0.88	0.88

The gradient boosting classifier, with a higher cross-validation score of 0.925 as shown in Table 7, can learn and adapt to data more effectively than the decision tree; however, its test accuracy matches that of the decision tree at 0.88. This suggests that despite its strength in training, the model's ability to generalize to unseen data is not superior.

This is similar to traffic on Ikorodu Road during non-peak hours, such as 7:00 a.m. or 12:00 p.m., when, while traffic is heavy, the flow does not improve significantly. Similarly to the gradient boosting model, traffic volumes fluctuate but do not show drastic improvements in efficiency during these times.

Table 7. Gradient boosting classifier.

Class	Precision	Recall	F1-Score
High	1.00	1.00	1.00
Low	0.80	0.80	0.80
Medium	0.67	0.67	0.67
Average/Total	0.88	0.88	0.88

The random forest classifier stood out with a perfect test accuracy of 1.00 and a solid cross-validation score of 0.894 as shown in Table 8. It excels in handling complex and variable patterns, just as traffic on Ikorodu Road fluctuates throughout the day with periods of high congestion, such as 16:00 and 17:00, when cars, jeeps, and buses dominate the road. The model's ability to generalize well, even with unpredictable traffic peaks, makes it the most efficient in handling diverse scenarios, akin to how a well-designed traffic system might adapt to heavy road use during rush hour.

Table 8. Random forest classifier.

Class	Precision	Recall	F1-Score
High	1.00	1.00	1.00
Low	1.00	1.00	1.00
Medium	1.00	1.00	1.00
Average/Total	1.00	1.00	1.00

Machine learning models and traffic data reflect similar patterns of fluctuation and predictability. The decision tree works well for stable and predictable situations, such as how rush-hour traffic follows a consistent rhythm. The gradient boosting classifier adapts better during training but does not significantly outperform the decision tree on the test set, much like traffic that remains steady, but does not improve drastically during certain times. Finally, the random forest classifier performs exceptionally, accurately predicting even the most complex and fluctuating traffic conditions, similar to a traffic system that adjusts seamlessly to both regular and unpredictable peak times.

4.6. Model Validation

The models were validated on a test dataset in which the models were not observed during training. The random forest classifier again proved to be the most accurate model, correctly predicting the traffic category for every instance in the test dataset. This suggests that the random forest classifier is the most reliable for future predictions, given its ability to generalize well to new data. However, further validation is recommended, using a more extensive and diverse dataset to confirm the robustness and reliability of the model.

Overall, the models performed well in predicting traffic categories based on the traffic data provided. Future work could consider tuning the hyperparameters of the models or incorporating additional relevant features to further improve their performance.

4.7. Traffic Condition Predictions

The predictive models developed in this study, specifically the random forest classifier, have shown significant potential for predicting future traffic conditions on Ikorodu

Road (Jibowu section). This model offers a reliable tool for forecasting traffic volumes and categorizing them into “Low”, “Medium”, and “High”. The prediction of traffic conditions is pivotal for traffic management and planning. Accurate predictions can facilitate the implementation of dynamic traffic management strategies, enabling traffic control centers to take preventative measures during peak hours or in anticipation of increased traffic volumes.

For instance, in anticipation of “High” traffic volumes, authorities can implement measures such as congestion pricing, rerouting traffic through alternative routes, or optimizing traffic signal timings. This proactive approach can significantly mitigate traffic congestion, reduce travel time, and improve overall road safety. This study provides insightful results regarding the traffic conditions on Ikorodu Road. It identifies patterns in traffic volume in relation to the day of the week and time of day. These models have shown promising accuracy in predicting traffic volumes, which could be an essential tool for traffic management.

However, this study had some limitations. This study is based on historical traffic data from a specific location, Ikorodu Road (Jibowu section). Hence, the predictions of the model may not be generalizable to other locations with different traffic patterns or conditions. Moreover, the study does not account for certain external factors, such as weather conditions, road work, accidents, or special events, which can significantly impact traffic volume. Future studies should consider these factors to obtain more comprehensive and accurate predictions.

In addition, further research could investigate the use of more sophisticated machine learning techniques or deep learning models, which might capture more complex patterns in the data and provide more accurate predictions. Another area of interest could be the exploration of real-time traffic prediction, which could offer even more dynamic and responsive traffic management solutions.

5. Conclusions

This study focused on analyzing and predicting traffic conditions on Ikorodu Road in Lagos. The analysis involved the examination of historical traffic data, specifically focusing on daily and hourly traffic volumes. The prediction involved the use of machine learning models, including decision trees, gradient boosting, and random forest classifiers. Our findings revealed significant variations in traffic volume across different days of the week and times of day, indicating peak and off-peak periods.

However, while our models were effective for predicting traffic conditions based on historical data, they also highlighted the need for a more comprehensive approach that includes additional factors, such as weather conditions, road work, and special events, which could significantly impact traffic volume. Based on our findings, we recommend the following:

1. Implementation of dynamic traffic management: The Lagos Traffic Management Authority should consider using a random forest classifier model or similar predictive tools to facilitate dynamic traffic management strategies. This could involve taking proactive measures during predicted peak times to mitigate congestion.
2. Consideration of additional factors: Future studies and traffic prediction models should incorporate additional external factors, such as weather conditions and special events, for more accurate and comprehensive traffic predictions.
3. Exploration of advanced modeling techniques: Further research could explore the use of more advanced machine learning techniques or deep learning models that might better capture complex patterns in the data and provide more accurate predictions.

4. Development of real-time traffic prediction systems: As technology and data collection techniques evolve, there is potential for developing real-time traffic prediction systems. Such systems can provide even more dynamic and responsive traffic management solutions.

In conclusion, although this study provides valuable insights and tools for traffic management on Ikorodu Road, there is still room for improvement and exploration of more advanced techniques in the field of traffic volume prediction.

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Abbreviations

ITS	Intelligent transportation systems
ADT	Average daily traffic
AADT	Annual average daily traffic
AI	Artificial intelligence
RFC	Random forest classifier
DT	Decision tree
GBC	Gradient boosting classifier
ML	Machine learning
X_i	Features
F	Trained model

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